Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala

Computer Science & Engineering Department

MST SEMESTER EXAMINATION

B. E.

Course Code: UCS745

Course Name: Clouds & DevOps

Date: 25th July, 2025

Time: 3 Hours, M. Marks: 30

Faculty: Dr. Amrita Kaur

Note: Attempt all 5 questions. Attempt the subparts of the question at one place.

1	Develop a Shell Script program to check whether a given year is a leap year or not.	(7)
2	There is a public repository named **"DATA-REPO"** on the CIT Hub under the **"PROJECTS TAB"**. Create a file named **"report.txt"** in the folder located at path D:\MyWork\Reports. Explain the steps required to push this file to the repository "DATA-REPO", assuming that this is your first time interacting with the CIT Hub Git repository.	(8)
3	(a) Differentiate Distributed and Centralised Version Control System? (4 marks) (b) Discuss the Docker Architecture in detail. (3 marks)	(7)
4	Discuss the product build lifecycle followed by Maven. Explain the key phases involved in the Maven build process and their significance in building and managing a Java project.	(8)